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Faculty of Energy, Industry,
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Electrical Engineering Program

Simulation of Symmetrical and Unsymmetrical Faults in DIgSILENT PowerFactory

COURSE:
Power Systems II

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1. Theoretical Framework

1.1 Symmetrical Components

Although three-phase systems normally operate under balanced conditions, faults can produce unbalanced currents and voltages in the three phases [1]. Analysis of these conditions is more complex than the analysis of symmetrical faults. Symmetrical faults are comparatively straightforward to evaluate but occur less frequently in practice; unsymmetrical faults are more common and are most effectively analyzed using symmetrical components.

The symmetrical-component method, developed by Charles Legeyt Fortescue, is one of the most powerful tools for analyzing unbalanced polyphase circuits. Fortescue's theorem states that an unbalanced set of n related phasors can be decomposed into n balanced phasor systems, known as symmetrical components. In this representation, $n - 1$ phasor sets are spatially displaced, while the remaining set consists of phasors that are in phase [2].

According to [3], the method can be applied to electrical networks of any voltage level. A symmetrical-component calculation is required when:

- the voltage or current system is unbalanced;
- a single-line-to-ground, line-to-line, or double-line-to-ground short circuit is considered; or
- the network contains rotating machines or specially connected transformers, particularly grounded-neutral connections.

Symmetrical components comprise three balanced systems: positive sequence, negative sequence, and zero sequence.

Before these sequences are examined, the operator a must be defined. This operator rotates a phasor by 120° in the complex plane, as expressed in Equation (1).

$$a = e^{j120^\circ} = -\frac{1}{2} + j\frac{\sqrt{3}}{2} \quad (1)$$

Table 1 lists identities involving a that are useful in subsequent calculations.

The composition and decomposition matrices are also fundamental to symmetrical-component analysis. They transform phasors from the sequence domain to the phase domain and vice versa. Equations (2) and (3) give the composition matrix A and decomposition matrix A^{-1} , respectively.

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & a \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} \quad (2)$$

Table 1. Common identities involving the operator a

$a^4 = a = 1\angle 120^\circ$
$a^2 = 1\angle 240^\circ$
$a^3 = 1\angle 0^\circ$
$1 + a + a^2 = 0$
$1 - a = \sqrt{3}\angle -30^\circ$
$1 - a^2 = \sqrt{3}\angle +30^\circ$
$a^2 - a = \sqrt{3}\angle 270^\circ$
$ja = 1\angle 210^\circ$
$1 + a = -a^2 = 1\angle 60^\circ$
$1 + a^2 = -a = 1\angle -60^\circ$
$a + a^2 = -1 = 1\angle 180^\circ$

$$\begin{bmatrix} V_0 \\ V_1 \\ V_2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} \quad (3)$$

Figure 1 illustrates the composition process and shows how unbalanced phase vectors are obtained from the three sequence-vector sets.

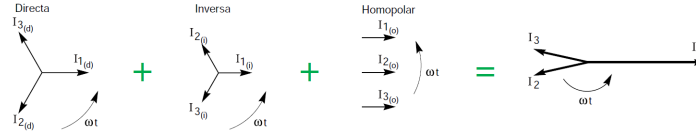


Figure 1. Graphical composition of positive-, negative-, and zero-sequence three-phase systems.

Source: Adapted from [3].

For short-circuit calculations, generators, motors, transformers, lines, and other network elements have specific parameters and configurations for each sequence. The complete power system is reduced to its sequence-network equivalents, as illustrated in Figure 2.

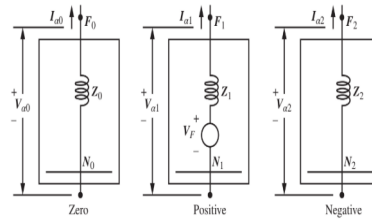


Figure 2. Reduced sequence networks.

Source: Adapted from [4].

All calculation methods considered in this study place a voltage source only in the positive-sequence network. This source represents the prefault voltage, whose value and treatment depend on the selected method or standard.

1.2 Faults in Electrical Power Systems

An electrical network is a dynamic system and is therefore exposed to abnormal conditions such as short circuits. These events increase current magnitudes and can impair protection performance and system stability [5]. Depending on the magnitude and relative phase of the voltage and current waveforms at fault inception, fault current may be asymmetric about the normal current axis. In general, maximum asymmetry occurs when the fault begins at a point on the voltage waveform other than its peak [6].

Once the fault is established, the current contains an alternating component (I_{AC}) and a direct component (I_{DC}), as occurs in AC RL circuits [6]. The DC component decays as its stored energy is dissipated as heat in the circuit resistance [7]. Its decay rate depends on the circuit X/R ratio: a lower X/R ratio produces faster decay. Figure 3 shows the decomposition of an asymmetric short-circuit current.

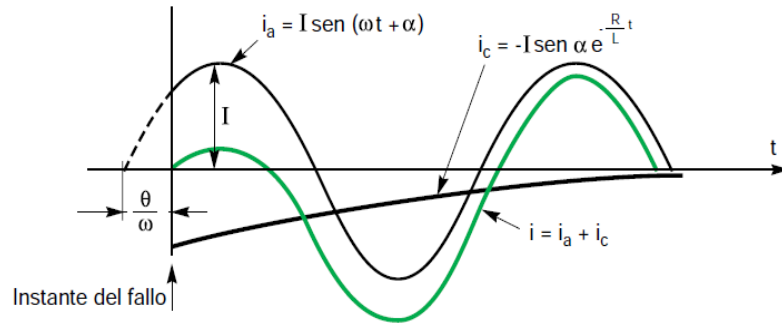


Figure 3. Components of an asymmetric short-circuit current.

1.2.1 Disturbances and Faults

Power-system abnormalities can be broadly classified, according to their severity, as disturbances or faults, as shown in Figure 4.

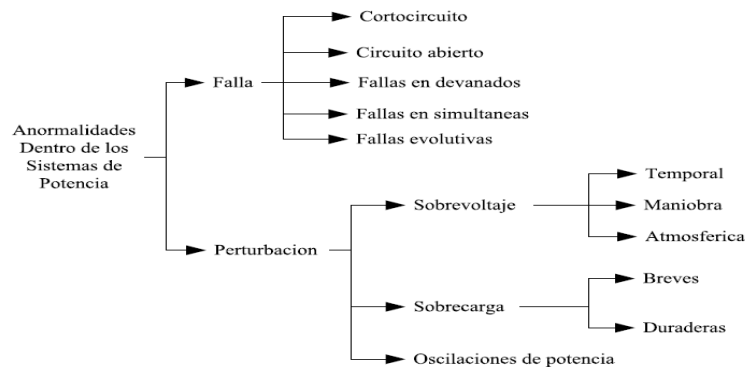


Figure 4. Abnormal conditions in an electrical power system.

Source: Adapted from [8].

Disturbances permit continued system operation but may damage equipment if they persist [8]. Because this study focuses on power-system faults, disturbances are not examined further.

A fault, in contrast, prevents one or more system components from operating normally and requires rapid protection-system intervention to limit damage.

Series faults occur when one or more conductors open without making contact with ground. Because no conductive connection is established between phases or ground, the condition is represented by an unbalanced series impedance [9, 10]. The unbalance is inserted in series with the line and therefore produces two fault terminals, one on each side of the discontinuity. The sequence network remains that of the otherwise symmetrical system, while the unbalanced portion is isolated outside the sequence network [4].

Simultaneous faults combine two or more faults occurring at the same time. They may have a common cause, independent causes, or a causal relationship. For example, an overhead-line conductor may break near a tower, remain suspended from an insulator on one side, and fall to ground on the other. Another example is a cross-country fault, in which faults occur on different phases and at different locations along a transmission line. Additional fault categories are discussed in [8]. The following subsection focuses on shunt faults.

1.2.2 Shunt Faults

Figure 5 classifies the principal shunt short circuits, which are the most common faults and the focus of this study.

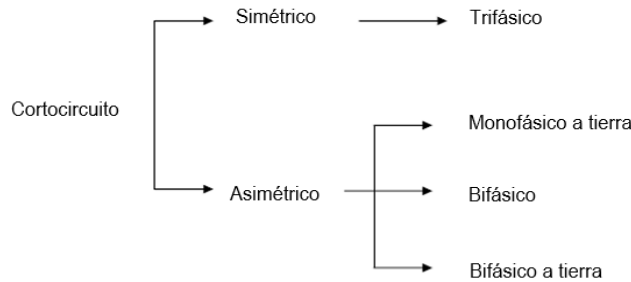


Figure 5. Classification of short-circuit faults.

Source: Adapted from [2].

Each fault type produces currents with characteristic magnitudes and phase relationships. Unsymmetrical faults are so named because their postfault phase currents do not have equal magnitudes separated by 120° .

1.2.2.1 Single-Line-to-Ground Short Circuit A single-line-to-ground fault connects one phase to ground. It is the most common short circuit, accounting for approximately 80% of occurrences [2]. It can also be severe, particularly in solidly grounded networks or networks grounded through low impedances [11]. Figure 6 presents its sequence-network representation.

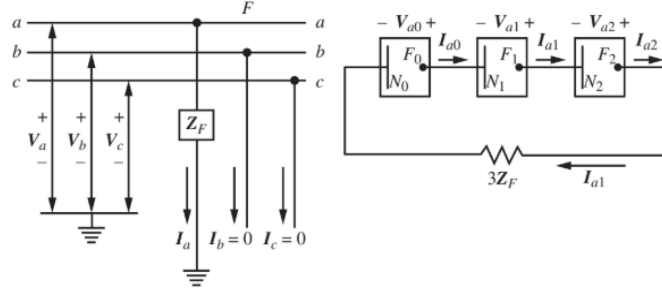


Figure 6. Single-line-to-ground fault.

Source: Adapted from [4].

Accurate calculation is important because the fault may produce high currents and ground-potential effects, including earth-leakage current, touch and step voltages, and electromagnetic interference.

All three sequence networks are required. As shown in Figure 6, they are connected in series, so the three sequence currents are equal. Equation (4) gives the resulting sequence current.

$$I_a^0 = I_a^1 = I_a^2 = \frac{V_F}{Z_{Th}^0 + Z_{Th}^1 + Z_{Th}^2 + 3Z_F} \quad (4)$$

Applying Equation (2) and the identities in Table 1 yields the faulted-phase current in Equation (5).

$$I_a = 3I_a^0 \quad (5)$$

1.2.2.2 Line-to-Line Short Circuit A line-to-line fault connects two phases. It produces unequal three-phase currents and requires the positive- and negative-sequence networks shown in Figure 7.

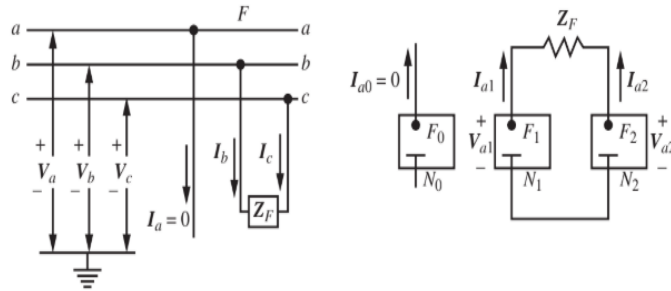


Figure 7. Line-to-line fault.

Source: Adapted from [4].

The initial symmetrical currents are generally lower than those of a three-phase fault. Near synchronous or induction machines, however, the resulting currents may become comparatively large [2].

Equations (6) and (7) describe the sequence currents. Because only two phases are involved and there is no ground connection, the reference-phase current (I_a) and zero-sequence current

(I_a^0) are zero.

$$I_a^1 = \frac{V_{Th}}{Z_{Th}^1 + Z_{Th}^2 + Z_F} \quad (6)$$

$$I_a^2 = -I_a^1 \quad (7)$$

Application of the composition matrix gives Equation (8).

$$I_b = -I_c = (I_a^1 \cdot a^2) + (I_a^2 \cdot a) = -j\sqrt{3}I_a^1 \quad (8)$$

1.2.2.3 Double-Line-to-Ground Short Circuit In a double-line-to-ground fault, two phases are connected to each other and to ground at the fault location. This fault has the lowest probability of occurrence among the shunt faults considered here. Because ground is involved, all three sequence networks must be included. [1] and [4] present two alternative formulations. The first places a single fault impedance between the joined phases and ground, as shown in Figure 8.

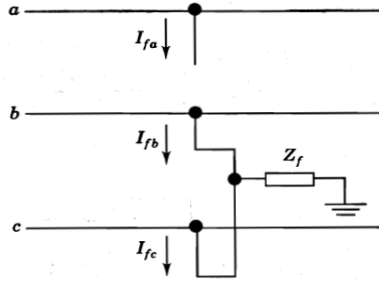


Figure 8. Double-line-to-ground fault: single ground-path impedance formulation.

Source: Adapted from [1].

The corresponding sequence currents are given by Equations (9)–(11).

$$I_a^1 = \frac{V_{Th}}{Z_{Th}^1 + [Z_{Th}^2 || (Z_{Th}^0 + 3Z_F)]} \quad (9)$$

$$I_a^2 = -I_a^1 \frac{Z_{Th}^0 + 3Z_F}{Z_{Th}^2 + (Z_{Th}^0 + 3Z_F)} \quad (10)$$

$$I_a^0 = -I_a^1 \frac{Z_{Th}^2}{Z_{Th}^2 + (Z_{Th}^0 + 3Z_F)} \quad (11)$$

The second formulation assigns a fault impedance to each sequence network, as shown in Figure 9 [4].

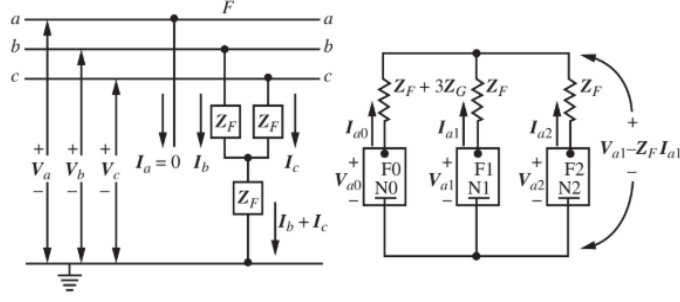


Figure 9. Double-line-to-ground fault: sequence-branch impedance formulation.
Source: Adapted from [4].

Equations (12)–(14) describe this formulation.

$$I_a^1 = \frac{V_{Th}}{(Z_{Th}^1 + Z_F) + [(Z_{Th}^2 + Z_F) \parallel (Z_{Th}^0 + Z_F)]} \quad (12)$$

$$I_a^2 = -I_a^1 \frac{Z_{Th}^0 + Z_F}{(Z_{Th}^2 + Z_F) + (Z_{Th}^0 + Z_F)} \quad (13)$$

$$I_a^0 = -I_a^1 \frac{Z_{Th}^2 + Z_F}{(Z_{Th}^2 + Z_F) + (Z_{Th}^0 + Z_F)} \quad (14)$$

1.2.2.4 Three-Phase Short Circuit A three-phase short circuit connects all three phases directly or through a low-valued impedance and generally imposes the greatest current duty on the system.

Because all phases are affected equally, the fault-point voltages become zero while the phase currents retain equal magnitudes separated by 120° . It is therefore the only symmetrical shunt fault [11]. Only the positive-sequence network is required, as shown in Figure 10.

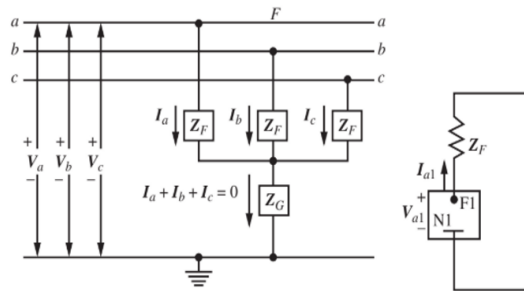


Figure 10. Three-phase fault.
Source: Adapted from [4].

Equations (15)–(17) describe the fault.

$$I_a^1 = \frac{V_{Th}}{Z_{Th}^1 + Z_F} \quad (15)$$

$$I_a^2 = I_a^0 = 0 \quad (16)$$

Applying the composition matrix gives:

$$I_a = I_b = I_c = I_a^1 \quad (17)$$

1.3 Short-Circuit Calculation Standards and Methods

Reliable short-circuit current estimates are required to define system modifications and equipment ratings that limit the consequences of faults. Typical applications include [2, 8, 12]:

- verification of the thermal withstand capability of system components;
- determination of the interrupting ratings of protective equipment;
- determination of a generating station's short-circuit capacity; and
- sizing of grounding and neutral equipment and design of substation grounding grids.

These applications require consistent and sufficiently accurate calculations. This study uses DIgSILENT PowerFactory [13] to compare ANSI, IEC 60909:2016, and the software's complete method.

1.3.1 ANSI Method

ANSI practice determines asymmetric fault currents by applying multiplying factors to symmetrical fault currents [2]. It evaluates separate resistance and reactance networks derived from the original system. Each network is reduced to a Thevenin equivalent, and the results are combined to obtain the final equivalent impedance [14]. For the momentary network, machine subtransient reactances are used to determine the current during the first fault cycles and derive quantities such as peak current.

All network elements are included in the equivalent impedance except motors rated below 50 hp. The multiplying factors account for the DC component and for the reduction of the AC component caused by changing machine reactances. These factors depend on the X/R ratio, calculated using Equation (18), where f and f_C are the system and equivalent-source frequencies, and X_C and R_C are the equivalent reactance and resistance.

$$\frac{X}{R} = \frac{X_C}{R_C} \cdot \frac{f}{f_C} \quad (18)$$

ANSI calculations commonly assume a prefault voltage V_F of 1.0 pu at every bus.

ANSI C37.010, the procedure considered in this study, addresses equipment applied in medium- and high-voltage systems. It distinguishes between local and remote generators, includes motor contributions, and provides the adjusted E/X method for applying circuit breakers on a symmetrical-current basis. This procedure considers three-phase and single-line-to-ground faults and uses a multiplying factor of 1.6 for momentary current and 1.0 for interrupting current when selecting breakers.

In PowerFactory, the ANSI calculation permits the user to specify V_F and options such as transformer tap treatment. It also provides a no-AC-decay (NACD) mode, whose factor characterizes the relative contribution of sources without AC decay. A contribution is treated as remote when the fault is electrically distant from the generator and local when the source is close to the fault.

NACD treatment depends on the selected option. The predominant method calculates the NACD factor and uses a DC-decay curve when the value is at least 0.5. The interpolated method corrects asymmetric fault currents through interpolation: at NACD equal to 1, only the DC-component factor is applied, whereas at NACD equal to 0, the combined AC/DC factor is used. The all-remote option sets NACD to 1 and uses a DC-decay curve; the all-local option sets NACD to 0 and uses an AC/DC decay curve. These options are directly related to the E/X procedure.

PowerFactory evaluates the following ANSI current duties [6, 14, 15]:

- **LV/Momentary.** Evaluates subtransient short-circuit current, including the symmetrical RMS momentary current and first-cycle equipment duty.
- **LV/Interrupting.** Evaluates the current used to determine the interrupting duty of medium- and high-voltage circuit breakers.
- **30 cycles.** Evaluates the steady-state current at 30 cycles and is used to determine minimum or sustained currents for current-operated protective relays.

The ANSI procedure can be summarized in four steps: identify the system elements, assemble their sequence impedances, reduce the network to a Thevenin equivalent at the fault location, and calculate the short-circuit currents.

PowerFactory reports the following quantities for an ANSI short-circuit calculation [16]:

- **Rated Voltage [kV].** Nominal voltage of the bus at which the study is performed.
- **Equivalent Impedance $R[\Omega]-X[\Omega]$.** Thevenin-equivalent impedance viewed from the fault location.
- **Symmetrical Current $[E/Z]$.** Symmetrical RMS short-circuit current for the selected duty, such as LV/Momentary or LV/Interrupting.
- **Apparent Power [MVA].** Short-circuit apparent power.
- **X/R ratio.** Ratio of equivalent reactance to resistance. A larger value increases the asymmetry factor and slows the decay of the DC component.
- **Asym. RMS X/R based [kA].** Asymmetrical RMS short-circuit current, calculated by applying an X/R -dependent correction factor to the symmetrical current.
- **Asym. Peak X/R based [kA].** Maximum instantaneous short-circuit current, including the DC component and its dependence on X/R .

1.3.2 IEC 60909:2016 Method

IEC 60909 applies to radial and meshed AC networks up to 230 kV. Based on Thevenin's theorem, it introduces an equivalent voltage source at the fault location and calculates the current at that point [2]. Network sources and synchronous and asynchronous machines are replaced by their sequence impedances.

Unlike the ANSI approach, IEC 60909 does not use decay curves to determine asymmetric current. Instead, it provides procedures for calculating peak current and the associated DC component.

The prefault voltage is represented through a voltage factor that accounts for expected voltage variation and, indirectly, the effects of transformer taps, network loading, transformer shunts, and rotating-machine subtransient behavior. Equation (19) defines the equivalent source voltage, and Figure 11 gives the standardized values of c .

$$V_F = \frac{c \cdot V_n}{\sqrt{3}} \quad (19)$$

TENSIÓN NOMINAL Vn	FACTOR DE TENSIÓN "c" PARA EL CÁLCULO DE	
	Icc máx.	Icc mín.
BT		
230 - 400 V	1	0,95
Otros	1,05	1
AT		
1 a 230 kV	1,1	1

Figure 11. IEC 60909 voltage factor c .

Source: Adapted from [2].

The IEC 60909 procedure comprises five principal steps: define the fault type, calculate the equivalent prefault voltage, reduce the applicable sequence networks to their Thevenin equivalents, calculate the initial short-circuit current, and derive quantities such as peak current.

The electrical distance between the fault and the generators must also be considered. For faults far from generators, where the AC component does not exhibit appreciable decay, the initial, interrupting, and steady-state short-circuit currents may be treated as equal. For generator-near faults, the steady-state current is lower than the interrupting current, which in turn is lower than the initial symmetrical current.

The initial symmetrical-current calculation follows the sequence-network equations presented for the shunt faults, with the IEC equivalent source voltage replacing the prefault voltage and with the appropriate treatment of generator proximity. Further details are provided in [2] and [11].

PowerFactory reports the following IEC 60909 quantities [16, 17]:

- **rtd.V**. Nominal voltage of the studied bus.
- **Voltage**. Phase voltages and angles during the fault.
- **c Factor**. Voltage factor defined in Figure 11; it represents voltage variation in space and time, possible transformer tap changes, and rotating-machine subtransient behavior.
- S_k'' [MVA]. Initial symmetrical three-phase short-circuit apparent power.
- I_k'' [kA]. Initial symmetrical short-circuit current, corresponding to the AC component at fault inception.
- i_p [kA]. Maximum possible instantaneous short-circuit current, including maximum asymmetry and therefore dependent on the circuit X/R ratio.
- I_b [kA]. Symmetrical breaking current used to rate and select circuit breakers; it is required primarily for generator-near faults.
- S_b [MVA]. Short-circuit breaking apparent power.
- **EFF**. Efficiency or contribution factor indicating the network contribution to short-circuit current. Values near 0 indicate a small contribution, whereas values near 1 indicate a substantial contribution.
- I_k [kA]. RMS steady-state short-circuit current after transient effects have decayed. Its value depends on generator magnetic-circuit saturation and is therefore less precise than I_k'' .
- I_{th} [kA]. Equivalent RMS current with the same thermal effect and duration as the actual short-circuit current; it is used to evaluate equipment thermal withstand.

These results support protective-device specification, protection coordination, and grounding-system design. [18] provides a more detailed treatment of IEC 60909 equations and distributed-generation contributions to short-circuit current.

1.3.3 Complete Method

PowerFactory's complete method applies superposition and incorporates the actual prefault network loading, which generally provides a more detailed result [16]. The calculation can be interpreted as the sum of a prefault circuit, obtained from a load-flow solution, and a fault-increment circuit [14].

Because the method uses the network impedance matrix and load-flow operating point, it can include line shunt admittances, loads, generator excitation conditions, transformer tap positions, the control options of the associated load-flow command, and the configured network models. It is therefore more comprehensive than methods based primarily on standardized correction factors.

The complete method is not restricted to balanced three-phase representations and can calculate short circuits in unbalanced network models. An unbalanced network or fault must, however, be initialized with an unbalanced load-flow calculation. The method can also calculate transient short-circuit currents when the required dynamic machine data are available [16].

Because it is a PowerFactory-specific method, it does not follow a single short-circuit standard and need not adopt assumptions that would reduce model detail. Nevertheless, some reported quantities are derived using concepts related to IEC 60909. These calculations are enhanced with the additional transient results available from the network model, and the output variables are therefore similar to those of the IEC method.

The user may initialize the complete method from a load-flow solution or apply a voltage correction factor c . If load-flow initialization is disabled, loads, shunt elements, and other components may be omitted through the calculation options. As in IEC 60909, the method provides procedures for deriving X/R -dependent quantities such as peak current, breaking current, and the DC component.

The reported results include initial symmetrical short-circuit power S''_k , initial symmetrical current I''_k , steady-state short-circuit current I_k , peak short-circuit current i_p , symmetrical breaking current I_b , contribution factor EFF, and equivalent thermal short-circuit current I_{th} . The complete method additionally reports the peak breaking current i_b .

2. Results

The following sections present the simulations and corresponding analyses for short circuits at Bus 4 of the proposed power system, shown in Appendix 1.

2.1 Three-Phase Short Circuit at Bus 4

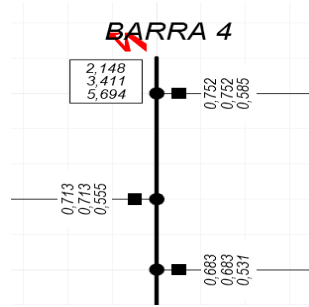


Figure 12. ANSI three-phase short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

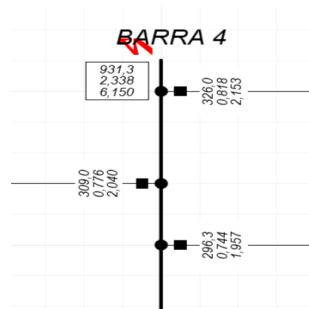


Figure 13. IEC 60909 three-phase short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

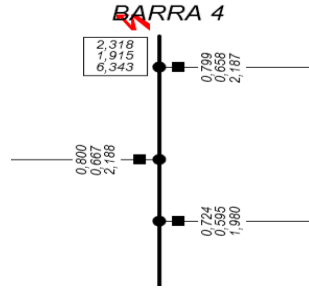


Figure 14. Complete-method three-phase short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

Table 2. Comparison of three-phase short-circuit calculation methods

Method	Symmetrical fault current [kA]	Short-circuit power [MVA]
ANSI standard	2.148	855.615
IEC 60909 standard	2.34	931.32
Complete method	2.32	923.47

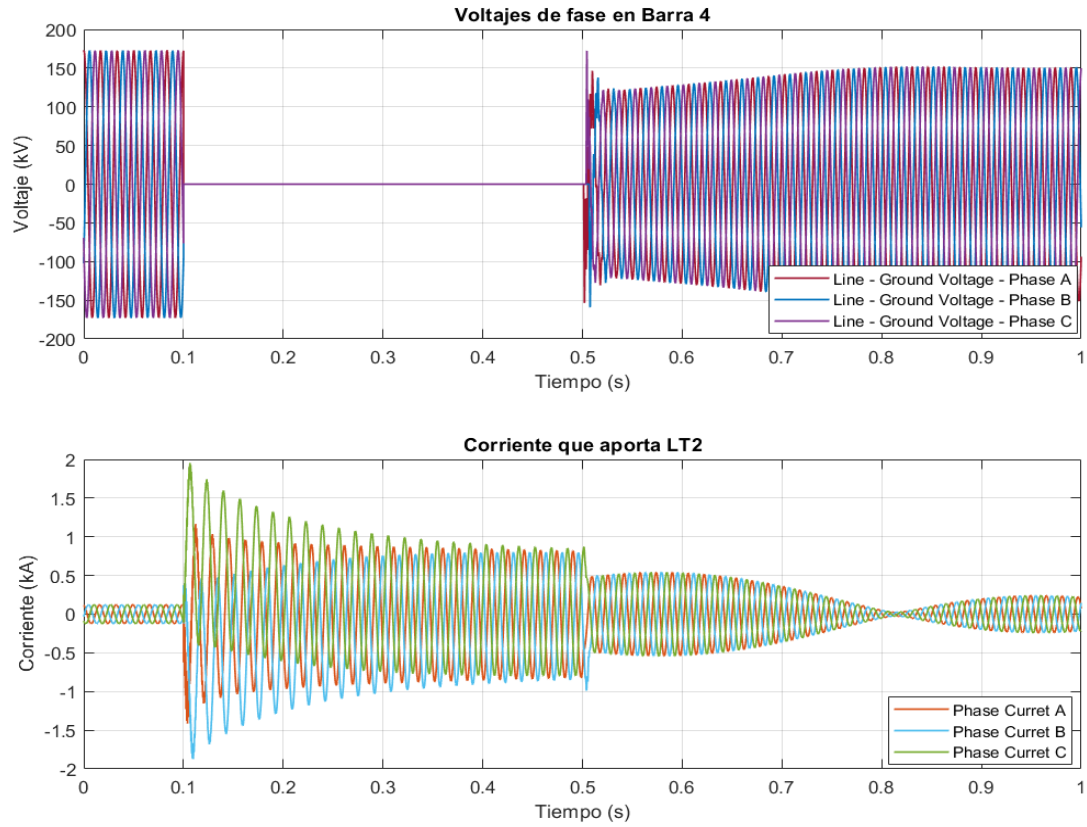


Figure 15. Results at Bus 4 and on LT2 for a three-phase short circuit in DIgSILENT PowerFactory.

Source: Author's work.

2.2 Single-Line-to-Ground Short Circuit at Bus 4

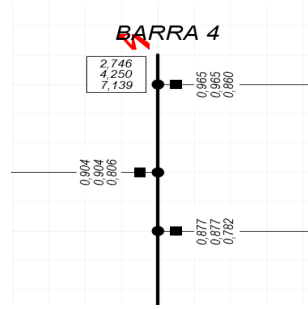


Figure 16. ANSI single-line-to-ground short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

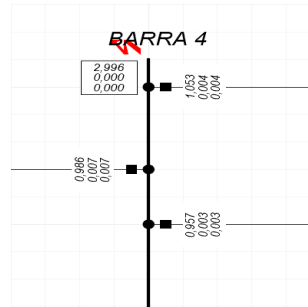


Figure 17. IEC 60909 single-line-to-ground short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

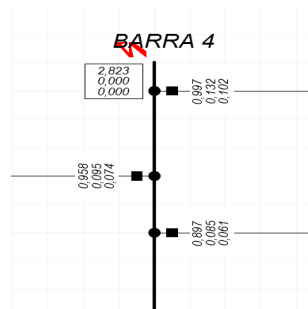


Figure 18. Complete-method single-line-to-ground short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

Table 3. Comparison of single-line-to-ground short-circuit calculation methods

Method	Symmetrical fault current [kA]	Short-circuit power [MVA]
ANSI standard	2.746	364.585
IEC 60909 standard	2.996	397.77
Complete method	2.82	374.97

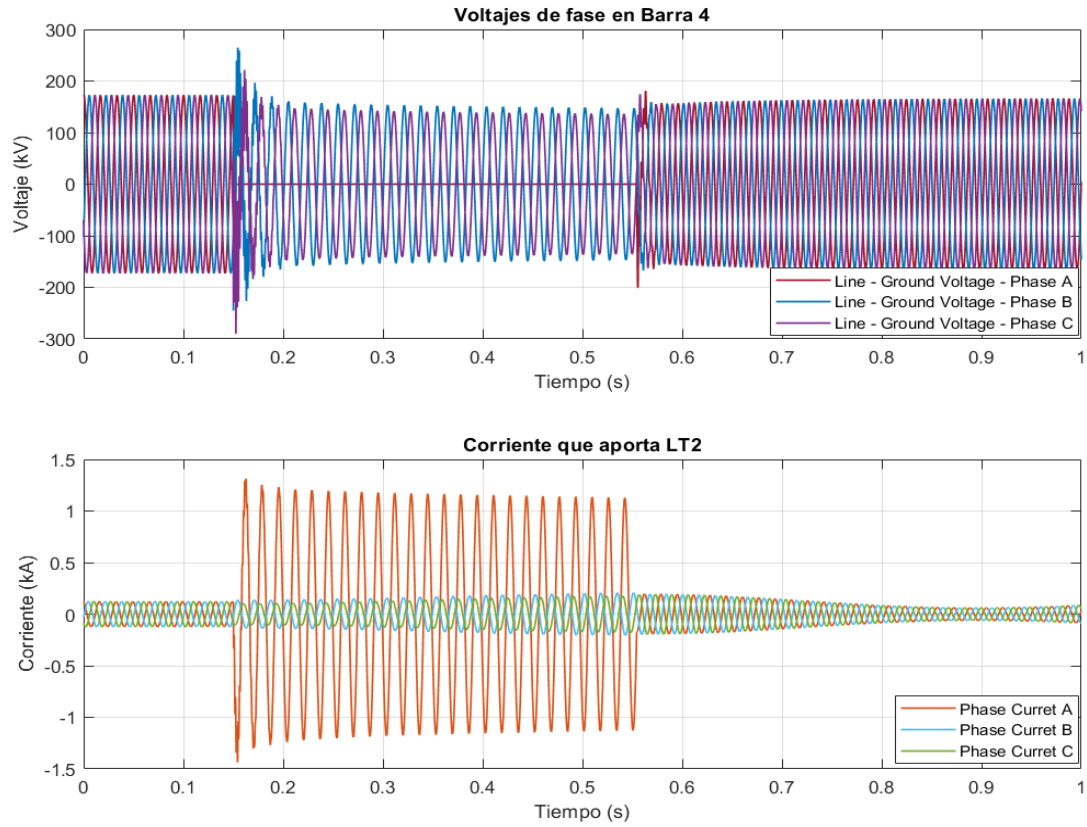


Figure 19. Results at Bus 4 and on LT2 for a single-line-to-ground short circuit in DIgSILENT PowerFactory.

Source: Author's work.

2.3 Line-to-Line Short Circuit at Bus 4

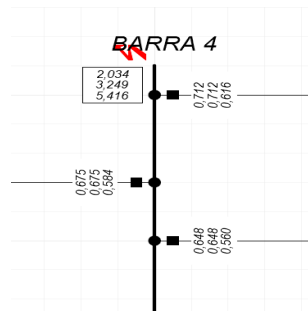


Figure 20. ANSI line-to-line short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

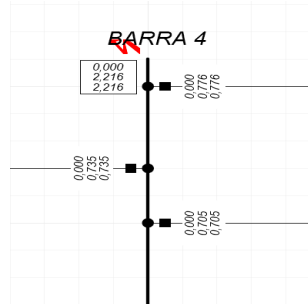


Figure 21. IEC 60909 line-to-line short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

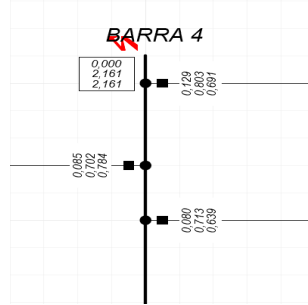


Figure 22. Complete-method line-to-line short circuit at Bus 4 in DIgSILENT PowerFactory.

Source: Author's work.

Table 4. Comparison of line-to-line short-circuit calculation methods

Method	Symmetrical fault current [kA]	Short-circuit power [MVA]
ANSI standard	2.034	270.162
IEC 60909 standard	2.22	294.22
Complete method	2.16	286.91

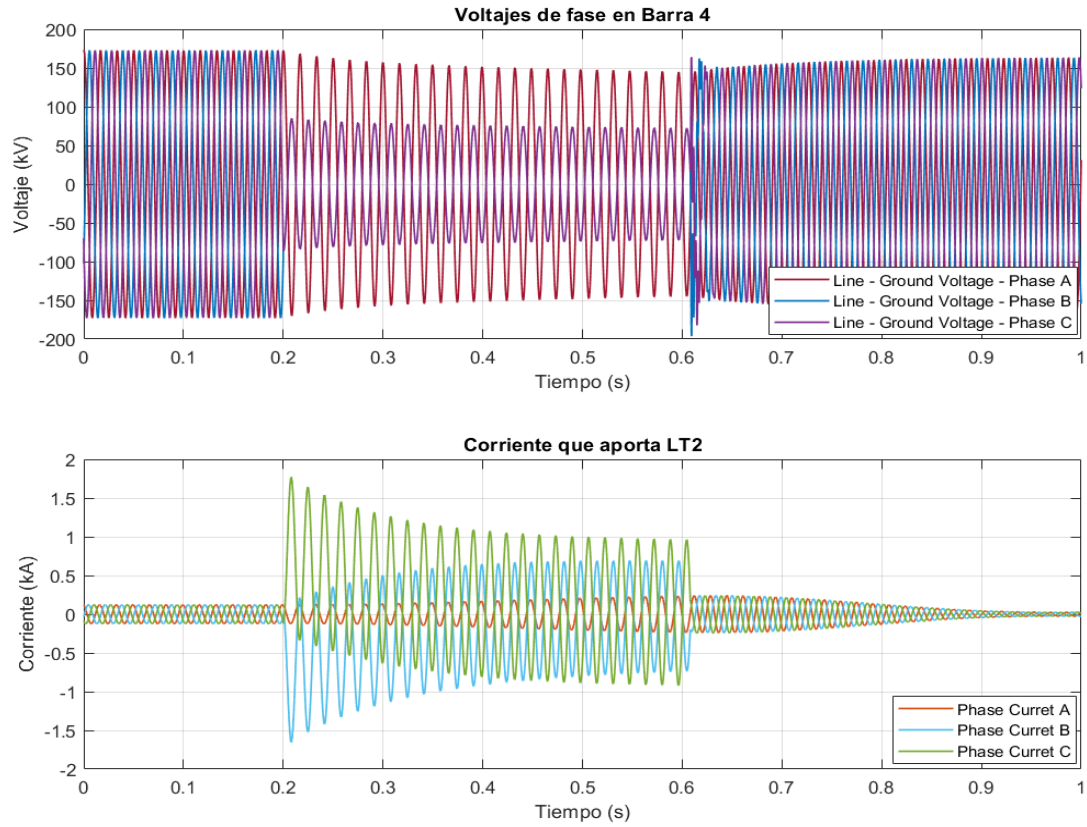


Figure 23. Results at Bus 4 and on LT2 for a line-to-line short circuit in DIgSILENT PowerFactory.
Source: Author's work.

2.4 Double-Line-to-Ground Short Circuit at Bus 4

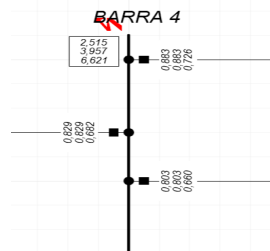


Figure 24. ANSI double-line-to-ground short circuit at Bus 4 in DIgSILENT PowerFactory.
Source: Author's work.

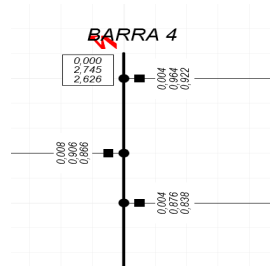


Figure 25. IEC 60909 double-line-to-ground short circuit at Bus 4 in DIgSILENT PowerFactory.
Source: Author's work.

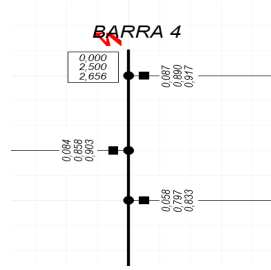


Figure 26. Complete-method double-line-to-ground short circuit at Bus 4 in DlgSILENT PowerFactory.

Source: Author's work.

Table 5. Comparison of double-line-to-ground short-circuit calculation methods

Method	Fault current B/C [kA]	Short-circuit power B/C [MVA]
ANSI standard	2.515 / 2.515	333.976 / 333.976
IEC 60909 standard	2.75 / 2.63	364.58 / 348.68
Complete method	2.50 / 2.66	331.98 / 352.63

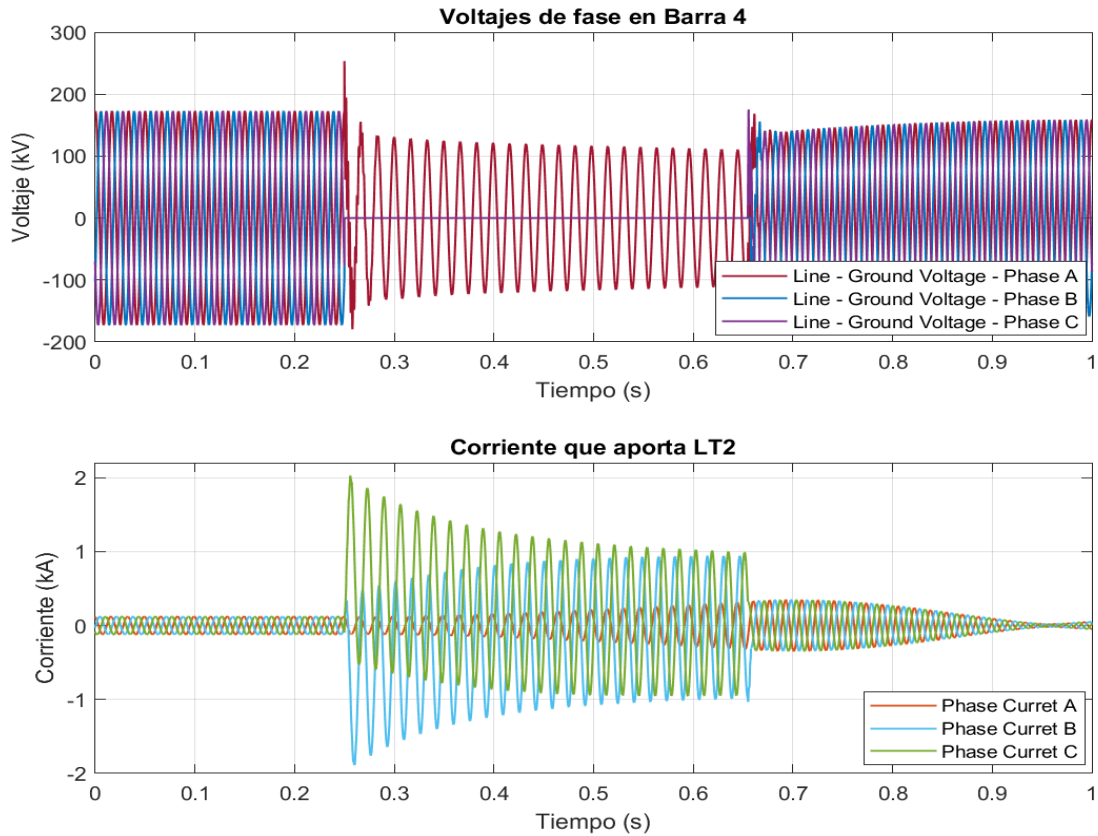


Figure 27. Results at Bus 4 and on LT2 for a double-line-to-ground short circuit in DlgSILENT PowerFactory.

Source: Author's work.

3. Discussion

The DIgSILENT PowerFactory analysis begins with a three-phase short circuit at Bus 4, evaluated using the three calculation methods. This fault provides the most straightforward case for comparison. The corresponding results are shown in Figures 12, 13, and 14; the short-circuit currents and powers extracted from the software reports are summarized in Table 2.

As discussed previously, the three methods apply different procedures and assumptions. The complete method provides the most detailed representation and is therefore used as the reference. Its results are close to those obtained with IEC 60909, although the methods use different prefault-voltage definitions. The same consideration applies when comparing the complete and ANSI methods. The discrepancies arise not only from prefault voltage but also from each method's internal correction factors and modeling assumptions. The simulations further show that Line 5 carries more current than Lines 3 and 4 because it is shorter and consequently has a lower impedance.

Overall, the contributions through LT2 and LT5 are similar under all three methods and are the largest line currents because these lines are close to the generators. This observation confirms that rotating machines provide the dominant contribution to fault current: Generator 1 supplies the fault through LT1 and LT2, whereas Generator 2 supplies it through LT5. Although the contribution from Generator 2 divides at Bus 6, the current reaching Bus 4 remains high because Generator 2 has the larger rating. An electromagnetic-transient (EMT) simulation was performed for a more detailed assessment, and its results are presented in Figure 15. This simulation does not require selection of a standardized short-circuit method; instead, it directly represents the time-domain response of the measured variables to the programmed event.

The three-phase fault was applied at $t = 0.1$ s and cleared at $t = 0.5$ s. Because all three phases are involved, the phase voltages at Bus 4 collapse to zero. Before the fault, the LT2 currents remain in steady state, with a peak of 0.1223 kA in phase C. Immediately after fault inception, the phase-C current rises to 1.956 kA.

The simulation also shows the decay of the DC component of fault current before clearance. The total current does not decrease substantially while the fault remains connected, despite the decay of its DC component. After $t = 0.5$ s, the current decreases; at approximately $t = 0.82$ s, however, its magnitude begins to oscillate and does not fully settle within the simulated interval.

The single-line-to-ground short circuit at Bus 4 was subsequently evaluated with the three methods. Figures 16, 17, and 18 show the results, and Table 3 lists the corresponding short-circuit currents and powers. IEC 60909 yields the highest phase-A fault current, 2.996 kA; ANSI yields the lowest, and the complete method gives an intermediate value. This ordering

is consistent with that observed for the three-phase fault.

For all three methods, the principal current contributions at Bus 4 originate from the generators along the shortest electrical paths. As in the three-phase case, the highest currents flow through LT1–LT2 and LT5, with LT5 generally carrying the largest value. This result reinforces the conclusion that Generator 2 makes the greatest contribution to each fault considered in this system.

Although the single-line-to-ground and three-phase faults exhibit similar contribution patterns, the single-line-to-ground currents are higher in this particular network. The explanation follows from the symmetrical-component formulation presented in Chapter 1. At first sight, inclusion of all three sequence impedances might be expected to reduce the single-line-to-ground current. For this fault, however, the three sequence currents are equal and the physical phase current is three times the sequence current. By contrast, the three-phase fault involves only the positive-sequence network. The three-phase short-circuit power is nevertheless higher because all three phases contribute. The EMT results for the single-line-to-ground fault are shown in Figure 19; the fault is applied at $t = 0.15$ s and cleared at $t = 0.55$ s.

Because only one phase is faulted, the phase-A voltage falls to zero during the event. Phases B and C initially exhibit transient peaks of 284.3 kV and 264.8 kV, respectively, and their voltages subsequently decrease until the fault is cleared, producing a pronounced voltage unbalance. Following clearance, another small transient occurs and the phase-A voltage recovers, although the postfault voltage remains below its prefault value because the system has not returned to its initial operating state. The LT2 phase currents show little visible attenuation of the DC component, unlike the three-phase case. This behavior indicates a comparatively high X/R ratio and, consequently, a slowly decaying DC offset.

The line-to-line short circuit at Bus 4 was then evaluated. The results are presented in Figures 20, 21, and 22, while Table 4 summarizes the corresponding currents and powers. For a solid line-to-line fault, the two faulted phases carry currents of equal magnitude, as confirmed by the results. Analysis of the line contributions again identifies LT2 and LT5 as the principal current paths. The previously observed ordering is also maintained: ANSI gives the minimum value, IEC 60909 the maximum, and the complete method an intermediate value.

Among the simulated faults, the line-to-line case produces the lowest short-circuit current and power. Only the positive- and negative-sequence networks participate in this fault. Application of the composition matrix produces a physical phase current equal to $\sqrt{3}$ times the sequence current, whereas the corresponding factor for a single-line-to-ground fault is 3. Combined with the higher equivalent impedance of the line-to-line sequence network, this difference explains the lower fault current.

The EMT simulation for this case is shown in Figure 23; the fault is applied at $t = 0.2$ s and cleared at $t = 0.6$ s. Because only phases B and C are involved, their phase-to-ground voltages

overlap during the fault. These voltages decrease but do not collapse directly to zero, while the phase-A voltage becomes unbalanced and decreases from its nominal value until clearance.

Additional simulations of line-to-line voltages show that V_{AB} and V_{CA} become unbalanced, whereas V_{BC} collapses to zero. Because the line-to-line fault has no direct connection to ground, each faulted phase retains a nonzero potential relative to ground, even though the potential difference between the two faulted phases is zero. In the previously analyzed three-phase and single-line-to-ground faults, the ground connection causes the corresponding phase-to-ground potential difference to collapse. The LT2 currents show a decaying DC component in both faulted phases. After clearance, the current remains near its nominal magnitude briefly and then decreases; over a longer simulation interval, residual oscillations indicate that the system is still returning toward its prefault operating condition.

Finally, the double-line-to-ground short circuit at Bus 4 was evaluated. Figures 24, 25, and 26 present the results, and Table 5 summarizes the corresponding currents and powers.

Unlike the line-to-line fault, this event connects two phases to ground; consequently, all three sequence networks participate, and the two faulted-phase currents need not have equal magnitudes. ANSI uses a more simplified representation and produces equal currents in both phases. IEC 60909 gives 2.75 kA in phase B and 2.63 kA in phase C, whereas the complete method gives 2.50 kA in phase B and 2.66 kA in phase C. Therefore, no single method produces the largest value in both phases. Nevertheless, the complete set of studies consistently shows that the largest contributions originate at the generators and reach the fault through LT2 and LT5.

Calculation of the double-line-to-ground current is more involved than for a line-to-line fault. The sequence currents are obtained by current division: the parallel combination of the zero- and negative-sequence impedances reduces the equivalent impedance, causing the positive-sequence current to be the largest sequence component. The sum of the zero- and negative-sequence currents equals the positive-sequence current in magnitude according to the adopted sign convention, and the physical phase currents are then obtained with the composition matrix. Because no single constant factor relates the sequence current to both faulted-phase currents, the magnitude cannot be ranked solely through a simple multiplier. In this network, the lower equivalent impedance can make the double-line-to-ground current greater than the line-to-line current, while the three-phase and single-line-to-ground cases may still produce larger values depending on the sequence impedances and prefault conditions.

The corresponding EMT simulation is shown in Figure 27; the fault is applied at $t = 0.25$ s and cleared at $t = 0.65$ s. At fault inception, phase A experiences an overvoltage while the voltages of the two grounded phases collapse to zero. The line-to-line voltages V_{AB} and V_{CA} decrease, whereas V_{BC} becomes zero.

The LT2 currents again exhibit decay of the DC component in both faulted phases. Their qualitative behavior resembles that of the line-to-line fault, but the magnitudes differ: phase C

reaches approximately 2 kA in the double-line-to-ground case, compared with approximately 1.7 kA for the line-to-line fault. In both cases, current oscillations remain after fault clearance.

4. Conclusions

- The results confirm that each short-circuit calculation method produces different values for the same scenario. ANSI generally yields the lowest values, whereas IEC 60909 produces the highest; the PowerFactory complete method, which uses the full network model rather than relying primarily on standardized assumptions, gives intermediate results. This finding shows that standard-specific assumptions—including the treatment of pre-fault voltage and parameter correction factors—produce conservative estimates that are useful for protection setting and coordination. Nevertheless, the complete method provides a more realistic representation because it incorporates prefault operating conditions and system dynamics.
- Generator proximity and rated capacity significantly affect short-circuit studies and fault-current magnitudes. In every case, lines LT1–LT2 and LT5 carried the highest short-circuit currents because they form the most direct paths between the generators and Bus 4. Generator 2, whose rated capacity is twice that of Generator 1, had the greatest influence on fault-current magnitude, demonstrating the direct relationship between machine rating and fault severity.
- In terms of current severity, the single-line-to-ground fault produced the highest peak short-circuit current, whereas the three-phase fault yielded the highest short-circuit apparent power because all three phases were involved. The line-to-line fault produced the lowest currents, consistently with symmetrical-component theory, which predicts a higher equivalent impedance when only the positive- and negative-sequence networks are combined.
- Several simulations exhibited the DC component of fault current and its dependence on the X/R ratio. A higher X/R ratio causes the DC component to persist longer. This behavior was particularly evident in the EMT simulation of the single-line-to-ground fault, for which the decay was less pronounced than for the other fault types and the fault current was also the highest. These results underscore the importance of the X/R ratio when assessing short-circuit magnitudes and their effects.
- In every time-domain simulation, once the fault was cleared, the phase voltages returned to a balanced state at a level below their prefault value. Currents in LT2 and LT5 exhibited damped oscillations that did not fully settle, even when the simulation interval was extended. Thus, although the fault was cleared, the system continued to exhibit its dynamic effects. In practice, fault clearance without appropriate protection or control adjustments may leave residual oscillations and potentially compromise system stability.

5. Recommendations

- Although this study summarizes the principal characteristics and differences among the short-circuit calculation methods available in DIgSILENT PowerFactory, the cited references should be examined in detail. Each source provides a more comprehensive treatment of the methods' theoretical foundations and applications, which is essential for interpreting the results in their proper context.
- A sound understanding of Thevenin-equivalent reduction and symmetrical components is fundamental to short-circuit analysis. Although PowerFactory facilitates the calculations, the principles underlying positive-, negative-, and zero-sequence networks, together with their application to different fault types, should be reviewed carefully.
- Manual short-circuit calculations are useful for consolidating theoretical knowledge and validating simple cases. For complex systems or studies involving numerous scenarios, however, manual calculations become impractical. Computational tools such as PowerFactory are therefore recommended because they support several short-circuit calculation methods and time-domain simulations of fault effects on system variables.
- The simulation of multiple time-domain fault scenarios can become repetitive and computationally demanding. PowerFactory automation capabilities, including the DIgSILENT Programming Language (DPL) and Python integration, should therefore be explored to streamline the workflow and facilitate systematic result analysis.
- Short-circuit analysis is fundamental to evaluating power-system dynamic behavior and provides a basis for more advanced studies, including contingency and stability assessments. Comprehensive proficiency requires rigorous theoretical preparation combined with systematic manual and software-assisted practice under different fault types and operating conditions.
- Future work should extend beyond fault-current calculation to examine system response strategies. Protection coordination is particularly important because properly designed and coordinated protection schemes limit fault consequences and help preserve system stability.

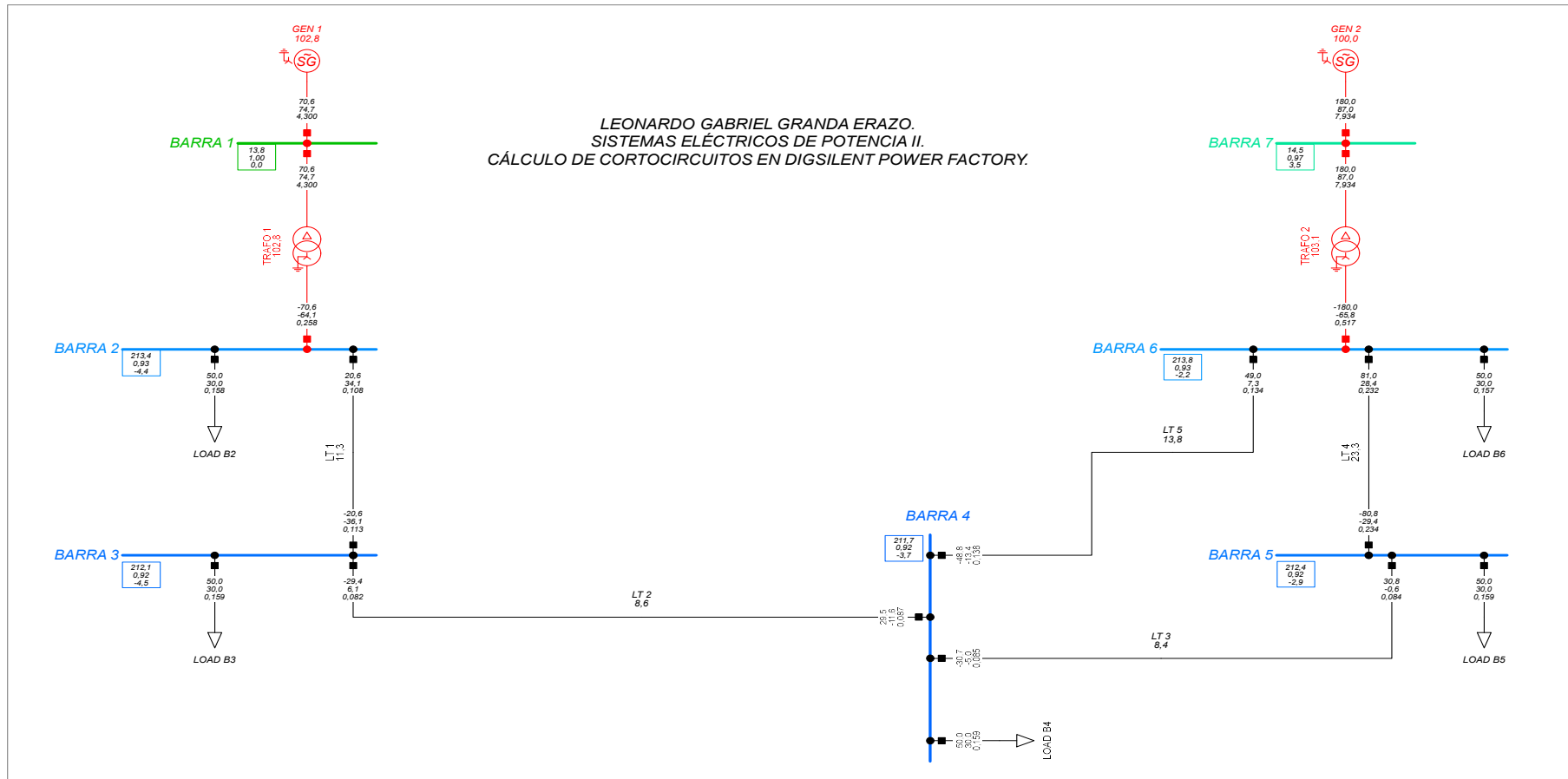
6. References

- [1] J. Grainger and W. Stevenson, *Análisis de sistemas de potencia*. McGraw-Hill, 2000, ISBN: 9789701009086.
- [2] L. A. V. VILLACIS, “Análisis de cortocircuitos a nivel de 230 kv y 138 kv con la segunda línea de interconexión entre ecuador y colombia y nueva generación,” *Escuela Politécnica Nacional*, 2007.
- [3] B. de Metz-Noblat, F. Dumas, and G. Thomasset, “Cálculo de corrientes de cortocircuito. Cuaderno Técnico Nº 158,” Schneider Electric España S.A., Barcelona, Tech. Rep., Sep. 2000, Original francés: noviembre 1999; versión española: septiembre 2000. [Online]. Available: <http://www.schneiderelectric.es>
- [4] P. M. Anderson, *Power System Protection*. Hoboken, NJ: Wiley-IEEE Press, 2020.
- [5] Y. N. Lafta, N. A. Shalash, Y. N. Abd, and A. A. Al-Lami, “Iec 60909 and ansi standards comparison with ascc based fault calculations of iraqi power system,” *Cogent Engineering*, vol. 6, no. 1, p. 1 705 654, 2019.
- [6] R. Ramos and A. B. Berrocal, “Metodologías para el calculo de las corrientes de cortocircuito,” *Universidad Tecnológica de Bolívar, Colombia*, 2011.
- [7] D. Gómez Marcial, “Análisis de contingencias eléctricas en centros comerciales,” Tech. Rep., El documento proporcionado corresponde al "Capítulo IV: Análisis de Corto Circuito".
- [8] F. Gonzalez-Longatt, *Sistemas de potencia 1: Capítulo 4: Anormalidades en sistemas de potencia 2007*, Oct. 2007. DOI: 10.13140/RG.2.2.32504.05124
- [9] D. P. Acuña Correa, “Localización de fallas en sistemas eléctricos de distribución basados en la matriz de impedancia de barra,” B.S. thesis, Universidad Politécnica Salesiana. Carrera de Ingeniería Eléctrica. Sede Quito, 2017.
- [10] J. J. M. Flórez, *Localización de faltas en sistemas de distribución de energía eléctrica usando métodos basados en el modelo y métodos basados en el conocimiento*. Universitat de Girona, 2007.
- [11] Á. J. I. C. C. Gómez and Á. L. Loaiza, “Estudio comparativo de las normas iec 60909-0 y ansi/ieee 141-4 para el cálculo de corrientes de cortocircuito en instalaciones eléctricas de uso final empleando los simuladores digsilent powerfactory y etap,” *Universidad Tecnológica de Pereira, Pereira*, 2017.
- [12] C. A. Chucos Huamán, *Digsilent powerfactory nivel intermedio: Estudios de cortocircuito y coordinación de protecciones*, Material del curso, CCITEC Ingeniería y Tecnología y Colegio de Ingenieros del Perú.
- [13] DIgSILENT GmbH, *PowerFactory 2022*, Software para análisis de sistemas eléctricos, Gomaringen, Alemania, 2022.

- [14] C. W. Aguirre Cárdenas, “Estudios eléctricos de sistemas de potencia para la carrera de ingeniería eléctrica utilizando el software power factory 13.1 de digsilent,” B.S. thesis, QUITO/EPN/2008, 2008.
- [15] *Análisis de cortocircuito*, Manual técnico para el análisis de cortocircuito, basado en la versión 14 de PowerFactory, describiendo métodos como IEC 60909, ANSI C37, y el Método Completo., PowerFactory.
- [16] DIgSILENT GmbH, *DIgSILENT PowerFactory User Manual 2022*, Versión 2022.1, DIgSILENT GmbH, Göttingen, Germany, May 2022. [Online]. Available: <https://www.digsilent.de>
- [17] International Electrotechnical Commission, “Short-circuit currents in three-phase a.c. systems – Part 0: Calculation of currents,” International Electrotechnical Commission, Geneva, Switzerland, Tech. Rep. IEC 60909-0:2016, Jan. 2016, © 2016 IEC, Geneva :contentReference[oaicite:0]index=0.
- [18] T. N. Boutsika and S. A. Papathanassiou, “Short-circuit calculations in networks with distributed generation,” *Electric Power Systems Research*, vol. 78, no. 7, pp. 1181–1191, 2008.

7. Appendices

Appendix 1. Power system under study



Appendix 2. ANSI single-line-to-ground short-circuit report

				DigSILENT PowerFactory 2022		Project: Date: 6/6/2025			
Fault Locations with Feeders -- Complete Report -- Short-Circuit Calculation / Method : ANSI						Single Phase to Ground			
Pre-fault Voltage Consider Transformer Taps		1,00 p.u. No	Fault Impedance Resistance, Rf Reactance, Xf		0,00 Ohm 0,00 Ohm	NACD Mode Currents/Voltages for	Interpolated LV/Interrupting		
Grid: Grid		System Stage: Grid				Annex:	/ 1		
	Rated Voltage [kV]	Equivalent Impedance R[Ohm] X[Ohm]	Symmetrical Current (E/Z) [kA] [deg]	Apparent Power [MVA]	X/R ratio	Asym.RMS X/R based [kA]	Asym.Peak X/R based [kA]	Sym.Base [kA]	Tot.Base [kA]
BARRA 4	230,00								
	Mom.Duty	2,688 61,768	2,746 -86,73	364,585	17,497	4,250	7,139		
	Zero-Seq	3,693 31,903					2 cycles	2,746	3,860
	Neg.-Seq	1,903 51,188					3 cycles	2,746	3,303
	Int.Duty	2,688 61,768	2,746 -86,73	364,585	17,497		5 cycles	2,746	3,026
	Zero-Seq	3,693 31,903					8 cycles	2,828	2,831
	Neg.-Seq	1,903 51,188							
	30-cycle	3,693 31,903	2,448 -86,94	325,089					
	Zero-Seq	3,693 31,903							
	Neg.-Seq	1,903 51,188							
LT 2									
	Mom.Duty	0,904 93,36	0,904 93,36	120,008		4,250	7,139		
	Int.Duty	0,904 93,36	0,904 93,36	120,008			2 cycles	2,746	3,860
	30-cycle	0,806 93,14	0,806 93,14	107,045			3 cycles	2,746	3,303
							5 cycles	2,746	3,026
							8 cycles	2,828	2,831
LT 3									
	Mom.Duty	0,877 93,23	0,877 93,23	116,465		4,250	7,139		
	Int.Duty	0,877 93,23	0,877 93,23	116,465			2 cycles	2,746	3,860
	30-cycle	0,782 93,02	0,782 93,02	103,830			3 cycles	2,746	3,303
							5 cycles	2,746	3,026
							8 cycles	2,828	2,831
LT 5									
	Mom.Duty	0,965 93,23	0,965 93,23	128,112		4,250	7,139		
	Int.Duty	0,965 93,23	0,965 93,23	128,112			2 cycles	2,746	3,860
	30-cycle	0,860 93,02	0,860 93,02	114,213			3 cycles	2,746	3,303
							5 cycles	2,746	3,026
							8 cycles	2,828	2,831
LOAD B4									
	Mom.Duty	0,000 0,00	0,000 0,00	0,000		4,250	7,139		
	Int.Duty	0,000 0,00	0,000 0,00	0,000			2 cycles	2,746	3,860
	30-cycle	0,000 0,00	0,000 0,00	0,000			3 cycles	2,746	3,303
							5 cycles	2,746	3,026
							8 cycles	2,828	2,831

Appendix 3. IEC 60909 single-line-to-ground short-circuit report

				DiGSILENT PowerFactory 2022			Project: ----- Date: 6/6/2025				
Fault Locations with Feeders											
Short-Circuit Calculation / Method : IEC 60909				Single Phase to Ground / Max. Short-Circuit Currents							
Asynchronous Motors Always Considered		Grid Identification Automatic				Short-Circuit Duration Break Time 0,10 s					
		Conductor Temperature User Defined No				Fault Clearing Time (Ith) 1,00 s					
						Voltage factor c Standard defined table					
Grid: Grid				System Stage: Grid				Annex: / 1			
		rtd.V. [kV]	Voltage [kV]	c- [deg] Factor	Sk" [MVA]	Ik" [kA]	[deg]	Ip [kA]	Ib [kA]	Sb [MVA]	EFF [-]
BARRA 4	A	230,00	0,00	0,00 1,10	397,77 MVA	3,00 kA	-86,80	7,88 kA	3,00	397,77	0,00
	B		123,84	-112,34	0,00 MVA	0,00 kA	0,00	0,00 kA	0,00	0,00	0,91
	C		129,48	112,30	0,00 MVA	0,00 kA	0,00	0,00 kA	0,00	0,00	0,93
LT 2	BARRA 3				A 130,94 MVA	0,99 kA	93,24	2,59 kA			
					B 0,98 MVA	0,01 kA	-84,55	0,02 kA			
					C 0,96 MVA	0,01 kA	-88,89	0,02 kA			
LT 3	BARRA 5				A 127,06 MVA	0,96 kA	93,19	2,52 kA			
					B 0,46 MVA	0,00 kA	95,45	0,01 kA			
					C 0,46 MVA	0,00 kA	91,11	0,01 kA			
LT 5	BARRA 6				A 139,77 MVA	1,05 kA	93,19	2,77 kA			
					B 0,51 MVA	0,00 kA	95,45	0,01 kA			
					C 0,50 MVA	0,00 kA	91,11	0,01 kA			
LOAD B4					A 0,00 MVA	0,00 kA	0,00	0,00 kA			
					B 0,00 MVA	0,00 kA	0,00	0,00 kA			
					C 0,00 MVA	0,00 kA	0,00	0,00 kA			

Appendix 4. Complete-method single-line-to-ground short-circuit report

										DIGSILENT PowerFactory 2022		Project: Date: 6/6/2025	
Fault Locations with Feeders													
Short-Circuit Calculation / Method : complete										Single Phase to Ground / Max. Short-Circuit Currents			
Short-Circuit Duration					Fault Impedance								
Break Time					0,10 s		Resistance, Rf			0,00 Ohm			
Fault Clearing Time (Ith)					1,00 s		Reactance, Xf			0,00 Ohm			
Grid: Grid		System Stage: Grid						Annex: / 1					
		rtd.V. [kV]	Voltage [kV]	c- [deg]	Factor	Sk* [MVA]	Ik* [kA]	Ik* [deg]	Ik' [kA]	ip [kA]	Ib [kA]	ib [kA]	EFF [-]
BARRA 4	A	230,00	0,00	0,00	1,00	374,87 MVA	2,82 kA	-81,8	2,61	7,72 kA	2,62	5,39	0,00
	B		112,89	-116,34		0,00 MVA	0,00 kA	0,0	0,00	0,00 kA	0,00	0,00	0,97
	C		106,28	112,39		0,00 MVA	0,00 kA	0,0	0,00	0,00 kA	0,00	0,00	0,90
LT 2	BARRA 3					A	127,24 MVA	0,96 kA	93,8	0,89	2,62 kA		
						B	12,62 MVA	0,10 kA	-95,3	0,09	0,26 kA		
						C	9,80 MVA	0,07 kA	142,7	0,07	0,20 kA		
LT 3	BARRA 5					A	119,18 MVA	0,90 kA	101,0	0,83	2,46 kA		
						B	11,31 MVA	0,09 kA	66,5	0,08	0,23 kA		
						C	8,04 MVA	0,06 kA	-82,8	0,06	0,17 kA		
LT 5	BARRA 6					A	132,38 MVA	1,00 kA	101,7	0,92	2,73 kA		
						B	17,48 MVA	0,13 kA	60,9	0,12	0,36 kA		
						C	13,58 MVA	0,10 kA	-92,1	0,10	0,28 kA		
LOAD B4						A	5,22 MVA	0,04 kA	-29,1	0,04	0,11 kA		
						B	17,65 MVA	0,13 kA	-132,2	0,12	0,36 kA		
						C	17,24 MVA	0,13 kA	65,0	0,12	0,36 kA		

Appendix 5. ANSI line-to-line short-circuit report

					DIGSILENT PowerFactory 2022		Project: Date: 6/6/2025			
Fault Locations with Feeders -- Complete Report --										
Short-Circuit Calculation / Method : ANSI					2-Phase Short-Circuit					
Pre-fault Voltage		1,00 p.u.		Fault Impedance						
Consider Transformer Taps		No		Resistance, Rf Reactance, Xf		0,00 Ohm 0,00 Ohm				
				NACD Mode		Interpolated				
				Currents/Voltages for		LV/Interrupting				
Grid: Grid		System Stage: Grid				Annex: / 1				
	Rated Voltage [kV]	Equivalent Impedance R[Ohm] X[Ohm]		Symmetrical Current (E/Z) [kA] [deg]		Apparent Power [MVA]	X/R ratio	Asym.RMS X/R based [kA]	Asym.Peak X/R based [kA]	
BARRA 4	230,00									
	Mom.Duty	2,688	61,768	2,034	-177,67	270,162	24,629	3,249	5,416	Sym.Base [kA]
	Zero-Seq	3,693	31,903						2 cycles	2,078
	Neg.-Seq	1,903	51,188						3 cycles	2,095
	Int.Duty	2,688	61,768	2,034	-177,67	270,162	24,629		5 cycles	2,034
	Zero-Seq	3,693	31,903						8 cycles	2,210
	Neg.-Seq	1,903	51,188							2,042
	30-cycle	3,693	31,903	1,760	-177,81	233,705				2,041
	Zero-Seq	3,693	31,903							
	Neg.-Seq	1,903	51,188							
LT 2	Mom.Duty			0,675	-177,55	89,593		3,249	5,416	
	Int.Duty			0,675	-177,55	89,593			2 cycles	2,078
	30-cycle			0,584	-177,69	77,552			3 cycles	2,095
									5 cycles	2,034
									8 cycles	2,210
										2,042
LT 3	Mom.Duty			0,648	2,28	85,994		3,249	5,416	
	Int.Duty			0,648	2,28	85,994			2 cycles	2,078
	30-cycle			0,560	2,16	74,374			3 cycles	2,095
									5 cycles	2,034
									8 cycles	2,210
										2,042
LT 5	Mom.Duty			0,712	2,28	94,594		3,249	5,416	
	Int.Duty			0,712	2,28	94,594			2 cycles	2,078
	30-cycle			0,616	2,16	81,811			3 cycles	2,095
									5 cycles	2,034
									8 cycles	2,210
										2,042
LOAD B4	Mom.Duty			0,000	0,00	0,000		3,249	5,416	
	Int.Duty			0,000	0,00	0,000			2 cycles	2,078
	30-cycle			0,000	0,00	0,000			3 cycles	2,095
									5 cycles	2,034
									8 cycles	2,210
										2,042

Appendix 6. IEC 60909 line-to-line short-circuit report

						DiGSILENT PowerFactory 2022		Project: Date: 6/6/2025					
Fault Locations with Feeders Short-Circuit Calculation / Method : IEC 60909												2-Phase Short-Circuit / Max. Short-Circuit Currents	
Asynchronous Motors Always Considered				Grid Identification Automatic				Short-Circuit Duration Break Time				0,10 s	
				Conductor Temperature User Defined				Fault Clearing Time (Ith)				1,00 s	
								Voltage factor c Standard defined table					
Grid: Grid				System Stage: Grid				Annex: / 1					
		rtd.V. [kV]	Voltage [kV]	c- [deg]	Factor	Sk" [MVA]	Ik" [kA]	[deg]	ip [kA]	Ib [kA]	Sb [MVA]	EFF [-]	
BARRA 4	A	230,00	132,31	0,75	1,10	0,00 MVA	0,00 kA	0,00	0,00 kA	0,00	0,00	1,00	
	B		66,15	-179,25		294,22 MVA	2,22 kA	-177,68	5,83 kA	2,22	294,22	0,00	
	C		66,15	-179,25		294,22 MVA	2,22 kA	2,32	5,83 kA	2,22	294,22	0,00	
LT 2	BARRA 3					A	0,06 MVA	0,00 kA	78,32	0,00 kA			
						B	97,56 MVA	0,73 kA	2,34	1,93 kA			
						C	97,57 MVA	0,73 kA	-177,63	1,93 kA			
LT 3	BARRA 5					A	0,03 MVA	0,00 kA	-101,68	0,00 kA			
						B	93,65 MVA	0,71 kA	2,31	1,86 kA			
						C	93,64 MVA	0,71 kA	-177,70	1,86 kA			
LT 5	BARRA 6					A	0,03 MVA	0,00 kA	-101,68	0,00 kA			
						B	103,01 MVA	0,78 kA	2,31	2,04 kA			
						C	103,01 MVA	0,78 kA	-177,70	2,04 kA			
LOAD B4						A	0,00 MVA	0,00 kA	0,00	0,00 kA			
						B	0,00 MVA	0,00 kA	0,00	0,00 kA			
						C	0,00 MVA	0,00 kA	0,00	0,00 kA			

Appendix 7. Complete-method line-to-line short-circuit report

												DigSILENT PowerFactory 2022		Project: Date: 6/6/2025									
Fault Locations with Feeders																							
Short-Circuit Calculation / Method : complete												2-Phase Short-Circuit				/ Max. Short-Circuit Currents							
Short-Circuit Duration						Fault Impedance																	
Break Time						Resistance, Rf						0,00 Ohm											
Fault Clearing Time (Ith)						Reactance, Xf						0,00 Ohm											
Grid: Grid				System Stage: Grid								Annex: / 1											
		rtd.V.		Voltage		c-		Sk"		Ik"		Ik'		ip		Ib		ib		EFF			
		[kV]		[kV]		[deg]		Factor		[MVA]		[kA]		[deg]		[kA]		[kA]		[kA]		[-]	
BARRA 4		A		230,00		112,94		-2,57		1,00		0,00 MVA		0,00 kA		0,0		0,00		0,00 kA		0,00	
		B		56,47		177,43				286,91 MVA		2,16 kA		-170,0		1,94		5,91 kA		1,96		4,06	
		C		56,47		177,43				286,91 MVA		2,16 kA		10,0		1,94		5,91 kA		1,96		4,06	
LT 2		BARRA 3						A		11,29 MVA		0,09 kA		20,4		0,08		0,23 kA					
								B		93,16 MVA		0,70 kA		4,4		0,63		1,92 kA					
								C		104,06 MVA		0,78 kA		-173,9		0,71		2,14 kA					
LT 3		BARRA 5						A		10,61 MVA		0,08 kA		169,8		0,07		0,22 kA					
								B		94,64 MVA		0,71 kA		11,0		0,64		1,95 kA					
								C		84,84 MVA		0,64 kA		-166,4		0,57		1,75 kA					
LT 5		BARRA 6						A		17,14 MVA		0,13 kA		163,2		0,12		0,35 kA					
								B		106,64 MVA		0,80 kA		10,3		0,72		2,20 kA					
								C		91,73 MVA		0,69 kA		-164,8		0,62		1,89 kA					
LOAD B4								A		19,52 MVA		0,15 kA		-33,5		0,13		0,40 kA					
								B		9,76 MVA		0,07 kA		146,5		0,07		0,20 kA					
								C		9,76 MVA		0,07 kA		146,5		0,07		0,20 kA					

Appendix 8. ANSI double-line-to-ground short-circuit report

				Digsilent PowerFactory 2022		Project: Date: 6/6/2025	
Fault Locations with Feeders -- Complete Report -- Short-Circuit Calculation / Method : ANSI						2-Phase to Ground	
Pre-fault Voltage Consider Transformer Taps		1,00 p.u. No	Fault Impedance Resistance, Rf Reactance, Xf		0,00 Ohm 0,00 Ohm	NACD Mode Currents/Voltages for	Interpolated LV/Interrupting
Grid: Grid		System Stage: Grid				Annex: / 1	
	Rated Voltage [kV]	Equivalent Impedance R[Ohm] X[Ohm]	Symmetrical Current (E/Z) [kA] [deg]	Apparent Power [MVA]	X/R ratio	Asym.RMS X/R based [kA]	Asym.Peak X/R based [kA]
BARRA 4	230,00						
	Mom.Duty	2,688 61,768	2,515 145,67	333,976	20,660	3,957	Sym.Base 6,621 [kA]
	Zero-Seq	3,693 31,903				2 cycles	2,547 3,627
	Neg.-Seq	1,903 51,188				3 cycles	2,578 3,134
	Int.Duty	2,688 61,768	2,515 145,67	333,976	20,660	5 cycles	2,579 2,852
	Zero-Seq	3,693 31,903				8 cycles	2,656 2,660
	Neg.-Seq	1,903 51,188					
	30-cycle	3,693 31,903	2,068 145,36	274,618			
	Zero-Seq	3,693 31,903					
	Neg.-Seq	1,903 51,188					
LT 2		Mom.Duty	0,829 -33,85	110,137		3,957	6,621
		Int.Duty	0,829 -33,85	110,137		2 cycles	2,547
		30-cycle	0,682 -34,18	90,627		3 cycles	2,578
						5 cycles	2,579
						8 cycles	2,656
LT 3		Mom.Duty	0,883 -34,56	106,593		3,957	6,621
		Int.Duty	0,883 -34,56	106,593		2 cycles	2,547
		30-cycle	0,660 -34,87	87,617		3 cycles	2,578
						5 cycles	2,579
						8 cycles	2,656
LT 5		Mom.Duty	0,883 -34,56	117,252		3,957	6,621
		Int.Duty	0,883 -34,56	117,252		2 cycles	2,547
		30-cycle	0,726 -34,87	96,379		3 cycles	2,578
						5 cycles	2,579
						8 cycles	2,656
LOAD B4		Mom.Duty	0,000 0,00	0,000		3,957	6,621
		Int.Duty	0,000 0,00	0,000		2 cycles	2,547
		30-cycle	0,000 0,00	0,000		3 cycles	2,578
						5 cycles	2,579
						8 cycles	2,656

Appendix 9. IEC 60909 double-line-to-ground short-circuit report

					DIGSILENT PowerFactory 2022		Project: Date: 6/6/2025				
Fault Locations with Feeders											
Short-Circuit Calculation / Method : IEC 60909					2-Phase to Ground			/ Max. Short-Circuit Currents			
Asynchronous Motors Always Considered				Grid Identification Automatic			Short-Circuit Duration Break Time 0,10 s Fault Clearing Time (Ith) 1,00 s				
				Conductor Temperature User Defined No			Voltage factor c Standard defined table				
Grid: Grid					System Stage: Grid			Annex: / 1			
		rtd.V. [kV]	Voltage [kV]	c- [deg] Factor	Sk" [MVA]	Ik" [kA]	[deg]	ip [kA]	Ib [kA]	Sb [MVA]	EFF [%]
BARRA 4	A	230,00	105,64	-1,19 1,10	0,00 MVA	0,00 kA	0,00	0,00 kA	0,00	0,00	0,76
	B		0,00	-120,00	364,58 MVA	2,75 kA	145,81	7,22 kA	2,75	364,58	0,00
	C		0,00	120,00	348,68 MVA	2,63 kA	41,17	6,91 kA	2,63	348,68	0,00
LT 2	BARRA 3				A 1,11 MVA	0,01 kA	94,44	0,02 kA			
					B 120,26 MVA	0,91 kA	-33,77	2,38 kA			
					C 115,84 MVA	0,87 kA	-139,22	2,28 kA			
LT 3	BARRA 5				A 0,53 MVA	0,00 kA	-85,56	0,01 kA			
					B 116,34 MVA	0,88 kA	-34,40	2,30 kA			
					C 111,26 MVA	0,84 kA	-138,64	2,20 kA			
LT 5	BARRA 6				A 0,58 MVA	0,00 kA	-85,56	0,01 kA			
					B 127,98 MVA	0,96 kA	-34,40	2,54 kA			
					C 122,38 MVA	0,92 kA	-138,64	2,42 kA			
LOAD B4					A 0,00 MVA	0,00 kA	0,00	0,00 kA			
					B 0,00 MVA	0,00 kA	0,00	0,00 kA			
					C 0,00 MVA	0,00 kA	0,00	0,00 kA			

Appendix 10. Complete-method double-line-to-ground short-circuit report

													DIGSILENT PowerFactory 2022		Project: Date: 6/6/2025												
Fault Locations with Feeders																											
Short-Circuit Calculation / Method : complete													2-Phase to Ground		/ Max. Short-Circuit Currents												
Short-Circuit Duration													Fault Impedance														
Break Time		0,10 s		Resistance, Rf		0,00 Ohm																					
Fault Clearing Time (Ith)		1,00 s		Reactance, Xf		0,00 Ohm																					
Grid: Grid			System Stage: Grid								Annex:				/ 1												
		rtd.V. [kV]		Voltage [kV]		c- [deg]		Factor		Sk* [MVA]		Ik* [kA]		Ik' [kA]		ip [kA]		Ib [kA]		ib [kA]		EFF [-]					
BARRA 4		A		230,00		96,51		0,54		1,00		0,00 MVA		0,00 kA		0,0		0,00		0,00 kA		0,00		0,00		0,83	
		B				0,00		-120,00				331,98 MVA		2,50 kA		153,7		2,17		6,84 kA		2,19		4,59		0,00	
		C				0,00		120,00				352,63 MVA		2,66 kA		44,9		2,30		7,27 kA		2,33		4,88		0,00	
LT 2		BARRA 3		A		11,09 MVA		0,08 kA		30,4		0,08		0,23 kA													
LT 3		BARRA 5		A		7,65 MVA		0,06 kA		-174,2		0,05		0,16 kA													
LT 5		BARRA 6		A		11,57 MVA		0,09 kA		176,1		0,08		0,24 kA													
LOAD B4				A		11,12 MVA		0,08 kA		-30,4		0,07		0,23 kA													
				B		5,56 MVA		0,04 kA		149,6		0,04		0,11 kA													
				C		5,56 MVA		0,04 kA		149,6		0,04		0,11 kA													

Appendix 11. ANSI three-phase short-circuit report

						DIGSILENT PowerFactory 2022		Project: ----- Date: 6/6/2025	
Fault Locations with Feeders -- Complete Report --									
Short-Circuit Calculation / Method : ANSI						3-Phase Short-Circuit			
Pre-fault Voltage		1,00 p.u.		Fault Impedance		0,00 Ohm		NACD Mode	
Consider Transformer Taps		No		Resistance, Rf		0,00 Ohm		Interpolated	
						Reactance, Xf		Currents/Voltages for	
								LV/Interrupting	
Grid: Grid						System Stage: Grid			Annex: / 1
		Rated Voltage	Equivalent Impedance	Symmetrical Current (E/Z)		Apparent Power	X/R ratio	Asym.RMS X/R based	Asym.Peak X/R based
		[kV]	R[Ohm] X[Ohm]	[kA] [deg]		[MVA]		[kA]	[kA]
BARRA 4		230,00							
		Mom.Duty	2,688 61,768	2,148	-87,51	855,615	22,993	3,411	5,694
		Int.Duty	2,688 61,768	2,148	-87,51	855,615	22,993		2 cycles
		30-cycle	3,081 79,402	1,671	-87,78	665,730			3 cycles
									5 cycles
									8 cycles
LT 2		Mom.Duty	0,713 92,58	283,853	22,161	3,411	5,694		
		Int.Duty	0,713 92,58	283,853	22,161				2 cycles
		30-cycle	0,555 92,29	221,093					3 cycles
									5 cycles
									8 cycles
LT 3		Mom.Duty	0,683 92,45	272,268	23,406	3,411	5,694		
		Int.Duty	0,683 92,45	272,268	23,406				2 cycles
		30-cycle	0,531 92,19	211,732					3 cycles
									5 cycles
									8 cycles
LT 5		Mom.Duty	0,752 92,45	299,495	23,406	3,411	5,694		
		Int.Duty	0,752 92,45	299,495	23,406				2 cycles
		30-cycle	0,585 92,19	232,906					3 cycles
									5 cycles
									8 cycles

Appendix 12. IEC 60909 three-phase short-circuit report

				DigSILENT		Project:						
				PowerFactory								
				2022		Date: 6/6/2025						
Fault Locations with Feeders												
Short-Circuit Calculation / Method : IEC 60909				3-Phase Short-Circuit / Max. Short-Circuit Currents								
Asynchronous Motors			Grid Identification			Short-Circuit Duration						
Always Considered			Automatic			Break Time 0,10 s						
						Fault Clearing Time (Ith) 1,00 s						
Decaying Aperiodic Component (idc)			Conductor Temperature			Voltage factor c						
Using Method B			User Defined No			Standard defined table						
Grid: Grid		System Stage: Grid				Annex: / 1						
	rtd.V.	Voltage	c-	Sk"	Ik"	ip	Ib	Sb	Ik	Ith		
	[kV]	[kV]	[deg]	Factor	[MVA]	[kA]	[deg]	[kA]	[MVA]	[kA]		
BARRA 4	230,00	0,00	0,00	1,10	931,32 MVA	2,34 kA	-87,06	6,15 kA	2,19	870,94	2,19	2,40
LT 2	BARRA 3				308,99 MVA	0,78 kA	92,97	2,04 kA				
LT 3	BARRA 5				296,35 MVA	0,74 kA	92,93	1,96 kA				
LT 5	BARRA 6				325,98 MVA	0,82 kA	92,93	2,15 kA				

Appendix 13. Complete-method three-phase short-circuit report

										DIGSILENT	Project:					
										PowerFactory						
										2022	Date: 6/6/2025					
Fault Locations with Feeders																
Short-Circuit Calculation / Method : complete										3-Phase Short-Circuit / Max. Short-Circuit Currents						
Short-Circuit Duration						Fault Impedance										
Break Time		0,10 s		Resistance, Rf		0,00 Ohm										
Fault Clearing Time (Ith)		1,00 s		Reactance, Xf		0,00 Ohm										
Grid: Grid				System Stage: Grid								Annex:		/ 1		
		rtd.V.	Voltage	c-	Sk"	Ik"		Ik'		ip	Ib	ib	Ith			
		[kV]	[kV]	[deg]	Factor	[MVA]	[kA]	[deg]	[kA]	[deg]	[kA]	[kA]	[kA]	[kA]		
BARRA 4		230,00	0,00	0,00	1,00	923,47 MVA	2,32 kA	-79,0	1,92	-75,7	6,34 kA	1,95	4,13	2,46		
LT 2		BARRA 3				318,60 MVA	0,80 kA	96,0	0,67	98,4	2,19 kA					
LT 3		BARRA 5				288,29 MVA	0,72 kA	103,4	0,60	107,3	1,98 kA					
LT 5		BARRA 6				318,42 MVA	0,80 kA	103,7	0,66	107,6	2,19 kA					
LOAD B4						0,00 MVA	0,00 kA	0,0	0,00	0,0	0,00 kA					